

100 Days of Linux - Day 066

Title: Making Decisions with if, else, and elif

Today's Goal

Learn how to make Bash scripts take different actions based on conditions using if, else, and elif.

Why This Matters

Decision-making is the foundation of automation. Production scripts often check whether files exist, validate user input, or determine whether a command succeeded before continuing.

Concepts of the Day

Basic Syntax:

```
if [ condition ]
then
commands
else
commands
fi
```

Examples:

```
if [ $AGE -ge 18 ]; then echo "Adult"; fi
if [ -f file.txt ]; then echo "File exists"; fi
if [ "$NAME" = "Vishal" ]; then echo "Welcome"; else echo "Hello"; fi
```

Common Tests:

-f (file exists), -d (directory exists), -eq, -ne, -gt, -lt, =, !=

Beginner Lesson

A condition is simply a true-or-false question. If the condition is true, Bash executes one block of commands. Otherwise, it executes another block.

Practice Questions

1. Create a script that asks for a user's age and prints 'Adult' if the age is 18 or greater; otherwise print 'Minor'.
2. Create a script that checks whether a file named notes.txt exists and prints an appropriate message.
3. Create a script that checks whether a directory named projects exists.
4. Modify a script to accept a name as a command-line argument and print 'Welcome Admin' only if the name is admin; otherwise print 'Welcome User'.
5. Run your scripts and display your current working directory.

Hints

Every if statement must end with fi. Leave spaces inside the square brackets: [condition].

Mini Challenge

Create login_check.sh that asks for a username and compares it

with 'admin'. If it matches, print 'Access Granted'; otherwise print 'Access Denied'. Then extend it with an elif branch for 'guest'.

Common Mistakes

Forgetting spaces around square brackets or forgetting the closing fi.

Did You Know?

Most deployment and backup scripts rely on if statements to verify prerequisites before making changes.

Pro Tip

Test conditions carefully before running commands that modify or delete files.

Answer Key (Instructor Only)

Q1: if [\$AGE -ge 18]; then echo Adult; else echo Minor; fi

Q2: if [-f notes.txt]; then echo 'File exists'; else echo 'File not found'; fi

Q3: if [-d projects]; then echo 'Directory exists'; else echo 'Directory not found'; fi

Q4: if ["\$1" = "admin"]; then echo 'Welcome Admin'; else echo 'Welcome User'; fi

Q5: bash script.sh && pwd

Homework

Create three scripts using numeric, string, and file existence conditions until writing if statements feels natural.

Next Day Preview

Tomorrow you'll learn loops with for and while to repeat tasks automatically.